CS383 · SOFTWARE ENGINEERING

Lecture 8 Requirement Engineering

Midterm summary

Muharib

What's inside

01 What is Requirement Engineering?

02 Feasibility study

03 Elicitation and analysis: four stages

04 Why is elicitation difficult?

05 Requirements discovery: four techniques

06 Scenario and use case guidelines

07 Requirements validation

08 Common mistakes

09 Test yourself

01 What is Requirement Engineering?

In Lecture 3 we saw that the first activity of building software is **Specification**, also called **Requirement engineering**. This lecture goes into its details: how we collect requirements, organize them, and make sure they are right.

The three stages

1. Requirements elicitation
2. Requirements analysis and specification
3. Requirements validation

Stage	What we do
Elicitation	Collect the requirements from the **Stakeholders**.
Analysis and specification	Organize, analyze and document them.
Validation	Check that they define the system the customer really wants.

02 Feasibility study

A short, focused study that takes place **early** in the requirement engineering process. It

answers three key questions:

#	Question
1	Does the system contribute to the overall objectives of the organization, and is it economically justifiable?
2	Can the system be implemented within schedule and budget using current technology?
3	Can the system be integrated with other systems that are used?

The rule

If the answer to **any** of these questions is no, then we probably do not go ahead with the project.

03 Elicitation and analysis: four stages

Technical staff work with clients to find the services the system should provide. The process

has four stages:

#	Stage	What happens
1	**Requirement discovery**	Interact with stakeholders to discover their requirements: interviews, scenarios, use cases, ethnography.
2	**Classification and organisation**	Organise the requirements into groups. An architectural design may be used to link requirements to components.
3	**Prioritization and negotiation**	Prioritize the requirements, and find and resolve conflicts between stakeholders.
4	**Requirements specification**	Document the requirements; they feed into the next round.

Watch out

Resolving conflicts between stakeholders happens in stage 3 (**Prioritization and negotiation**).

04 Why is elicitation difficult?

Reason	Meaning
They often don't know what they want	Stakeholders know what they want only in the most general terms.
They express requirements in their own terms	Based on their own knowledge and experience.
Different stakeholders, different requirements	Engineers must discover all sources, and find commonalities and conflicts.
Political factors	Managers may demand requirements that increase their influence in the organization.
Dynamic economic and business environment	New requirements may come from new stakeholders who were not consulted at first.

05 Requirements discovery: four techniques

#	Technique	In short
1	**Interviewing**	Ask the stakeholders, formally or informally.
2	**Scenarios**	A real-life example of how to interact with the system.
3	**Use cases**	Identify the actors and draw the interaction in UML.
4	**Ethnography**	The analyst goes into the working environment and observes.

1. Interviewing

Type	Meaning
Closed interview	The stakeholder answers predefined questions.
Open interview	No predefined questions; there are issues to discuss for a better understanding.

An effective interviewer is:

- Open-minded, avoids preconceived ideas about the requirements, and is willing to listen.
- Prompts the interviewee to get the discussion going, to understand the context of the requirement.

2. Scenarios

A real-life example of how to interact with the system. It starts as an outline and is developed into a complete description. A scenario may include:

- What to expect when the scenario starts
- The normal flow of events
- What may go wrong
- Information about other activities
- The system state when the scenario finishes

Slide example: Buy a product

```
1. Customer browses catalogue and selects items
```

```
2. Customer goes to checkout
```

```
3. Customer fills in shipping information (new customer)
```

```
4. System presents full pricing information
```

```
5. Customer fills in credit card information
```

```
6. System authorises purchase
```

```
7. System confirms sale immediately
```

```
8. System sends confirmation email
```

```
Alternative 6.a: authorisation fails → re-enter card and retry
```

```
Alternative 3.a: regular customer → saved shipping info shown
```

Steps 1 to 8 are the normal flow; the alternatives cover what may go wrong and other paths.

3. Use cases

A method for capturing the **functional requirements**. It identifies the **actors** in a scenario, describes their interaction with the system, and is usually drawn in UML. It can also be used to design the user interface.

UML symbol	Meaning
Oval	**Use case**
Stick figure	**Actor**
Solid line	**Association**: links an actor to a use case
Dashed arrow <<include>>	**Include**: a use case always contains another one
Dashed arrow <<extend>>	**Extend**: adds behaviour in certain cases
Rectangle	**System boundaries**
Arrow with a triangle head	**Generalization**

Slide example (ATM): a customer uses an ATM connected to a bank to check the balance, withdraw cash and pay bills. Every operation must be approved by the bank, the ATM authenticates through the bank, and an observer watches every transaction.

Element	In the example
Actors	User, Transaction Observer
System boundaries	<<ATM>> and <<Bank>>
ATM use cases	check balance, withdraw cash, pay bills, handle invalid access, handle invalid operation, handle abort operation
Bank use cases	Authentication, Operation Approval

4. Ethnography

A complementary study of the social and organizational context of the work. The analyst gets deeply into the working environment and discovers **implicit** requirements that other techniques missed: how people actually work, not how the client describes the process.

06 Scenario and use case guidelines

1. Name the system scope and boundaries
2. Brainstorm and list the primary actors
3. Brainstorm and list user goals for the system
4. Capture the outermost summary use cases
5. Reconsider and revise the summary use cases
6. Double-check for time-based triggers and events
7. List the extensions and conditions
8. Write the main success scenario (MSS)
9. Write alternative scenarios

Start with the boundaries and the actors, then their goals, and finally write the main success scenario (**MSS**) and the alternatives.

07 Requirements validation

Checking that the requirements actually define the system the customer really wants.

The five checks

Check	The question it answers
Validity	Do the requirements serve the needs of all stakeholders?
Consistency	Are there conflicting requirements? Resolve them.
Completeness	Are all functions and constraints included? Nothing is missing.
Realism	Can the requirements actually be implemented?
Verifiability	Are they written so that a set of tests can be written for them?

The three validation techniques

Technique	Meaning
Requirements reviews	Check the requirements systematically for errors and inconsistencies.
Prototyping	An executable temporary version of the system shown to end-users and the customer.
Test-case generation	Write a test for each requirement. If a test is hard to write, the requirement is hard to test. In agile, tests are written from user requirements before coding.

08 Common mistakes

Mistake	Correct
Saying the feasibility study happens after design	It happens early, at the start of requirement engineering.
Mixing up Consistency and Completeness	Consistency: no conflicts. Completeness: nothing missing.
Mixing up Realism and Verifiability	Realism: it can be built. Verifiability: it can be tested.
Saying use cases capture non-functional requirements	They capture functional requirements.
Mixing up closed and open interviews	Closed: predefined questions. Open: discussion without predefined questions.
Listing ethnography as a validation technique	It is a discovery technique. Validation: reviews, prototyping, test-case generation.

09 Test yourself

1. What are the three stages of requirement engineering, in order?
2. In which stage of elicitation and analysis are conflicts between stakeholders resolved?
3. The analyst goes into the workplace and observes people at work. What is this technique called?
4. Which check: Ensure that requirements can actually be implemented.
5. Name two requirements validation techniques.

Answers

#	Answer
1	Elicitation, then Analysis and specification, then Validation.
2	**Prioritization and negotiation**
3	**Ethnography**
4	**Realism**
5	Two of: Requirements reviews, Prototyping, Test-case generation.